Noether-Lefschetz Loci, Monodromy Representations, and Degree Thresholds for Hypersurfaces in Lean 4

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/NoetherLefschetz
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23652976

Earliest Commit Provenance: 2026-09-21 04:24:19 AM CDT (UTC-5)
Siloed Archival Edition: October 25, 2026 (America/Chicago CDT)

Abstract

Standalone Lean 4 formalization (‘NoetherLefschetz/Basic.lean‘, ‘NoetherLefschetz/Spec.lean‘) and technical documentation (‘docs/MONODROMY.md‘, ‘docs/DEGREE_THRESHOLDS.md‘, ‘docs/COUNTEREXAMPLES_CONTEXT.md‘) of Noether-Lefschetz degree thresholds and generic Picard/Hodge rank restrictions for smooth projective hypersurfaces.

1 Scope, Scientific Honesty, and Non-Claim Boundary

This repository formalizes degree thresholds and Noether-Lefschetz structural specifications. It does not claim a proof or disproof of the Clay Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/NoetherLefschetz`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652976`
- **Earliest Git Commit SHA:** `64b4d7bb85ecc51e416ba5bc3a6b53a54bb735e3` (2026-09-21 04:24:19 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Author: Benjamin Stanley Frohman ([@BenFrohman](https://github.com/BenFrohman))
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This repository is **not** a proof of the Hodge conjecture.
It records the Noether–Lefschetz theorem as a separate statement.
On a *very general* hypersurface of high enough degree, there are **no extra Hodge classes**. The only Hodge classes are powers of the hyperplane class, which are already algebraic. Hodge is then true on that host for a trivial reason: there is nothing extra to construct.
Let \{}(S \{}subset \{}mathbb{P}^3\{}) be a very general surface of degree \{}(d \{}ge 4\{}).
Then

\{}[
\{}operatorname{Pic}(S) \{}cong \{}mathbb{Z}\{}cdot \{}mathcal{O}_S(1),
\{}qquad
\{}operatorname{Hdg}^1(S) = \{}mathbb{Q}\{}, h.
\{}]
Lefschetz (1921); modern proofs via Hodge theory / infinitesimal variation
(Griffiths, Carlson, Green, Voisin, …).

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/NoetherLefschetz` is audited via `titan-audit` (`audit/TITAN_AUDIT_CER` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	**Size**	**SHA-256 Prefix**
`COPYRIGHT`	399 B	`bcce150caf6575c8...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	328 B	`b8978c158d483435...`
`NoetherLefschetz/Basic.lean`	1088 B	`d80121b7a82becaf...`
`NoetherLefschetz/Spec.lean`	1530 B	`b52392e9842230b1...`
`README.md`	2572 B	`838f7cd67cd7c466...`
`docs/MONODROMY.md`	3839 B	`c45ce598d9af083e...`
`docs/STATEMENT.md`	953 B	`df1156d5d3d35cb4...`
`docs/THRESHOLDS.md`	1589 B	`ecc76ad529ea19ff...`
`docs/VACUOUS.md`	616 B	`47287ad9b88ed196...`
`docs/WHAT_COMPILES.md`	1441 B	`53cb757853f2f9f9...`

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